Finite-Time Nonsingular Terminal Sliding Mode Control for 3D Trajectory Tracking of a Quadrotor UAV under Exogenous Perturbations

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***Abstract*— This paper presents a nonsingular terminal sliding mode control (NTSMC) strategy for robust three-dimensional (3D) trajectory tracking of a quadrotor unmanned aerial vehicle (UAV) operating in the presence of external disturbances and modeling uncertainties. The proposed controller is designed using the full nonlinear dynamics of the quadrotor and ensures finite-time convergence of the sliding variables and tracking errors without introducing singularities in the control law. A Lyapunov-based stability analysis is provided to demonstrate finite-time stability of the closed-loop system. The effectiveness of the proposed approach is evaluated through comparative simulations against representative state-of-the-art sliding mode control strategies. Numerical simulations are conducted under multiple operating scenarios, including large initial deviations and persistent external perturbations. The results demonstrate that the proposed NTSMC scheme achieves improved tracking accuracy, enhanced robustness, and a more favorable transient response compared to the considered benchmark controllers.**

I. Introduction

Quadrotor unmanned aerial vehicles (UAVs) have undergone rapid technological development over the past decade and are increasingly deployed in a wide range of civilian, industrial, and defense-related applications. Their vertical take-off and landing capability, compact mechanical structure, and high maneuverability make them particularly suitable for tasks such as aerial inspection, surveillance, environmental monitoring, mapping, and autonomous navigation. Due to their capability to operate in both indoor and outdoor environments, quadrotor platforms have become a prominent research topic in modern aerial robotics [1]–[6].

From the perspective of control engineering, quadrotor UAVs represent nonlinear underactuated dynamical systems characterized by six degrees of freedom and only four independent control inputs corresponding to the total thrust and three control torques. Although this structure enables lightweight design and agile flight, it also introduces significant challenges in controller synthesis due to nonlinearities, strong coupling between translational and rotational dynamics, and the presence of external disturbances.

Early studies on quadrotor control relied primarily on conventional strategies such as proportional–integral–derivative (PID) control, linear–quadratic regulator (LQR) techniques, backstepping approaches, and classical sliding mode control (SMC), which enabled stable autonomous flight of micro aerial vehicles [7], [8]. With the advancement of aerial robotics research, these control strategies have been further refined to support increasingly demanding flight tasks and aggressive maneuvering [1], [9].

Due to its inherent robustness with respect to matched disturbances and model uncertainties, SMC has become one of the most widely investigated approaches for quadrotor stabilization and trajectory tracking. Numerous SMC-based control strategies have been proposed to enhance disturbance rejection and robustness properties [2], [4], [10]–[12]. Nevertheless, several limitations remain in the existing literature. Many results rely on simplified or reduced-order quadrotor models, thereby neglecting important coupling effects inherent in the full nonlinear dynamics. Moreover, finite-time stability is not always rigorously established for the complete system, particularly in the presence of disturbances. In addition, some approaches rely on restrictive initial conditions or implicitly eliminate the reaching phase, which may lead to an overly optimistic assessment of robustness and an underestimation of chattering effects. As a result, discrepancies between theoretical predictions and practical system behavior may arise.

In this context, classical first-order sliding mode control (FOSMC) methods provide reliable disturbance rejection. However, their discontinuous control structure typically produces the well-known chattering phenomenon, which may excite unmodeled dynamics and accelerate actuator degradation [9], [13]. To alleviate this drawback, higher-order sliding mode techniques have been introduced. In [14], a second-order sliding mode control (SOSMC) scheme was proposed for full-degree-of-freedom trajectory tracking of a quadrotor UAV using the complete nonlinear model. The developed controller ensured finite-time convergence of the tracking errors and demonstrated improved robustness compared with classical first-order sliding mode designs, while a high-gain adaptation (HGA) mechanism was introduced to mitigate chattering effects. Despite these advances, further improvements in chattering attenuation and transient performance are still required, particularly under disturbance-affected operation. These observations motivate the exploration of alternative sliding surface designs capable of enhancing convergence properties while preserving robustness.

In this work, an NTSMC strategy is developed for robust three-dimensional (3D) trajectory tracking of a quadrotor UAV. A Lyapunov-based analysis is provided to establish finite-time stability of the closed-loop system. The proposed design is formulated using the complete nonlinear six de-

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grees of freedom model and explicitly accounts for bounded external disturbances acting in both translational and rotational channels. This provides a more realistic assessment of robustness and control performance under practical operating conditions.

The main contributions of this paper can be summarized as follows:

- Development of a nonsingular terminal sliding mode control (NTSMC) framework for 3D trajectory tracking of a quadrotor UAV, formulated using the complete nonlinear six degrees of freedom model and explicitly accounting for coupling effects.
- Lyapunov-based finite-time stability analysis that incorporates modeling uncertainties and external disturbances, providing rigorous robustness guarantees.
- Comprehensive comparative evaluation with FOSMC and SOSMC approaches under realistic disturbance scenarios, highlighting improved robustness and transient performance of the proposed method.
- Analysis of the trade-off between chattering reduction and robustness through the incorporation of a HGA mechanism.

remainder ???

The remainder of the paper is organized as follows. Section II presents the nonlinear kinematic and dynamic model of the quadrotor UAV. Section III describes the proposed NTSMC control design and provides the corresponding stability analysis. Simulation results and comparative performance evaluation are reported in Section IV. Finally, concluding remarks are given in Section V.

II. Kinematic and Dynamic Model of the Quadrotor UAV

The quadrotor UAV is considered as a rigid-body aerial platform driven by four rotors placed symmetrically with respect to the vehicle center of mass at a distance l. Two propellers rotate clockwise, while the remaining two rotate counterclockwise, which ensures compensation of the reaction torques and preserves the overall yaw equilibrium. Each rotor generates an aerodynamic thrust proportional to the square of its angular velocity.

The vehicle motion is characterized by six degrees of freedom consisting of three translational and three rotational components. Two coordinate systems are introduced in order to describe the vehicle motion. The inertial reference frame $\{g\}$ represents the global coordinate system, whereas the body-fixed frame $\{o\}$ is attached to the center of mass of the quadrotor, as illustrated in Fig. 1. The position of the vehicle in the inertial frame is defined by the coordinates (x, y, z), while the orientation is described using Euler angles roll ϕ, pitch θ, and yaw ψ. The linear velocities (u, v, w) and angular velocities (P, Q, R) are expressed in the body-fixed coordinate frame.

A. Forces and Moments

The thrust generated by each rotor can be expressed as

$$F_i = b\Omega_i^2, \quad i = 1, 2, 3, 4, \tag{1}$$

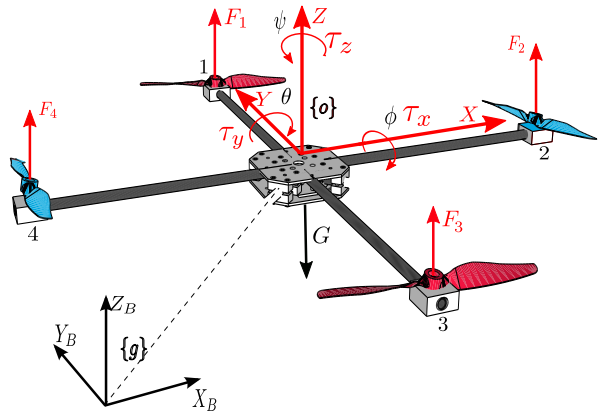

Fig. 1: Quadrotor UAV configuration showing thrust forces, control torques, and reference frames

where b denotes the thrust coefficient and Ω_i represents the angular velocity of the i-th rotor. The total thrust produced by the quadrotor is directed along the body-fixed z-axis and can be written as

$$T_z = F_1 + F_2 + F_3 + F_4 = b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2). \tag{2}$$

Besides the propulsive forces, the quadrotor is affected by gravity. In the inertial reference frame, the gravitational force vector is given by

$$\mathbf{F}_g^{\{g\}} = \begin{bmatrix} 0 & 0 & -mg \end{bmatrix}^T, \tag{3}$$

where m denotes the total mass of the vehicle and g is the gravitational acceleration constant. When this force is represented in the body-fixed frame, its components become dependent on the Euler angles, resulting in nonlinear coupling terms in the translational dynamics

$$\mathbf{F}_g^{\{o\}} = mg \begin{bmatrix} s_\theta & -s_\phi c_\theta & -c_\phi c_\theta \end{bmatrix}^T, \tag{4}$$

where $s_\alpha = \sin(\alpha)$ and $c_\alpha = \cos(\alpha)$.

The control torques generated by differential rotor speeds can be expressed as

$$\tau = \begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} bl(\Omega_1^2 - \Omega_3^2) \\ bl(\Omega_4^2 - \Omega_2^2) \\ d(\Omega_2^2 + \Omega_4^2 - \Omega_1^2 - \Omega_3^2) \end{bmatrix}, \tag{5}$$

where d represents the aerodynamic drag coefficient associated with the rotation of the rotor. The overall control input vector can therefore be written as

good model introduction

$$\mathbf{u}_c = \begin{bmatrix} T_z & \tau_x & \tau_y & \tau_z \end{bmatrix}^T. \tag{6}$$

B. Translational Kinematics and Dynamics

The kinematic relations describing the transformation between body-frame velocities and inertial position derivatives are given by .

$$\dot{x} = c_\theta c_\psi u + (s_\phi s_\theta c_\psi - c_\phi s_\psi)\, v + (c_\phi s_\theta c_\psi + s_\phi s_\psi)\, w, \tag{7}$$

$$\dot{y} = c_\theta s_\psi u + (s_\phi s_\theta s_\psi + c_\phi c_\psi)\, v + (c_\phi s_\theta s_\psi - s_\phi c_\psi)\, w, \tag{8}$$

$$\dot{z} = -s_\theta u + s_\phi c_\theta v + c_\phi c_\theta w. \tag{9}$$

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The translational dynamics expressed in the body-fixed frame can be written as

$$\dot{u} = Rv - Qw + gs_\theta, \tag{10}$$

$$\dot{v} = Pw - Ru - gs_\phi c_\theta, \tag{11}$$

$$\dot{w} = Qu - Pv - gc_\phi c_\theta + \frac{T_z}{m} + d_T, \tag{12}$$

where $d_T(t)$ represents a bounded disturbance affecting the thrust channel.

C. Rotational dynamics

The rotational dynamics follows the rigid-body Euler equations

$$\dot{P} = \frac{I_{yy} - I_{zz}}{I_{xx}} QR + \frac{\tau_x}{I_{xx}} - \frac{I_{zzm}}{I_{xx}} QW_G + d_{\tau_x}, \tag{13}$$

$$\dot{Q} = \frac{I_{zz} - I_{xx}}{I_{yy}} PR + \frac{\tau_y}{I_{yy}} - \frac{I_{zzm}}{I_{yy}} PW_G + d_{\tau_y}, \tag{14}$$

$$\dot{R} = \frac{I_{xx} - I_{yy}}{I_{zz}} PQ + \frac{\tau_z}{I_{zz}} + d_{\tau_z}, \tag{15}$$

where I_{xx}, I_{yy}, and I_{zz} denote the principal moments of inertia of the quadrotor body. The parameter I_{zzm} represents the rotor moment of inertia with respect to the rotor z-axis in its local reference frame, while d_{τ_i} denotes bounded disturbance torques acting on the rotational channels.

D. Euler Angle Kinematics

The relationship between body angular velocities and Euler angle rates is described by

$$\dot{\phi} = P + Qs_\phi t_\theta + Rc_\phi t_\theta, \tag{16}$$

$$\dot{\theta} = Qc_\phi - Rs_\phi, \tag{17}$$

$$\dot{\psi} = Q\frac{s_\phi}{c_\theta} + R\frac{c_\phi}{c_\theta}. \tag{18}$$

The above equations represent the complete nonlinear six degrees of freedom model of the quadrotor UAV. This formulation captures the coupling between translational and rotational dynamics, the influence of gravitational forces, and the nonlinear coordinate transformation between inertial and body-fixed frames. Additional details on quadrotor UAV modeling can be found in [15]. The obtained model serves as the basis for the controller design presented in the following section.

III. NTSMC Design for Robust 3D Trajectory Tracking of a Quadrotor UAV

In this section, the NTSMC technique is adopted in order to achieve finite-time convergence of the sliding variables and tracking errors while avoiding singularities typically associated with classical terminal SMC formulations. The NTSMC algorithm is designed for the four directly actuated variables of the quadrotor UAV. The control inputs (the total thrust and three control torques) directly affect the altitude z and attitude variables ϕ, θ and ψ, while the horizontal positions x and y are indirectly controlled through the flat mapping between translational and rotational dynamics [6], [13], [14].

A. NTSMC-Based Altitude Control Design

The objective of the altitude control subsystem is to ensure accurate tracking of the desired altitude z_{ref} by designing a nonsingular terminal SMC that guarantees finite-time convergence of the altitude tracking error.

The tracking error associated with the altitude dynamics is defined as

$$e_z = z - z_{ref}, \tag{19}$$

while the corresponding sliding surface is designed as

Sliding Surface; interesting term

$$\sigma_z = e_z + \frac{1}{c_{1z}} \dot{e}_z^{\frac{p}{q}} = 0, \tag{20}$$

where σ_z denotes the altitude sliding variable and $c_{1z} > 0$ is a design parameter. The integers p and q are selected as odd numbers satisfying $1 < \frac{p}{q} < 2$. The adopted nonsingular terminal sliding surface (20) ensures finite-time convergence of the tracking error while avoiding singularities that may arise in classical terminal sliding mode formulations.

The equivalent control term is obtained using the sliding condition $\dot{\sigma}_z = 0$, i.e.

$$\dot{\sigma}_z = \dot{e}_z + \frac{1}{c_{1z}} \frac{p}{q} \dot{e}_z^{(\frac{p}{q}-1)} \ddot{e}_z = 0. \tag{21}$$

Differentiating (19) twice and substituting the result into (21) yields

$$\ddot{z} = \ddot{z}_{ref} - c_{1z} \frac{q}{p} \dot{e}_z^{(2-\frac{p}{q})}. \tag{22}$$

Differentiating (9) gives

$$\ddot{z} = -c_\theta \dot{\theta} u - s_\theta \dot{u} + \dot{A}_1 v + A_1 \dot{v} + A_2 \dot{w} + \dot{A}_2 w, \tag{23}$$

where the auxiliary variables $A_1 = c_\theta s_\phi$ and $A_2 = c_\theta c_\phi$ are introduced for notational compactness.

By substituting the quadrotor dynamics into the sliding condition and equating (22) with (23), while accounting for (12), the equivalent control term is expressed as

$$T_{zeq} = \frac{m}{A_2} \Big[\ddot{z}_{ref} - c_{1z} \frac{q}{p} \dot{e}_z^{(2-\frac{p}{q})} + c_\theta \dot{\theta} u + s_\theta \dot{u} - \dot{A}_1 v - A_1 \dot{v} - \dot{A}_2 w - A_2 Q u + A_2 P v + g A_2^2 \Big], \tag{24}$$

To ensure convergence toward the sliding surface, the switching control term

$$T_{zconv} = -c_{2z} \operatorname{sgn}(\sigma_z), \tag{25}$$

is introduced, where $c_{2z} > 0$ is a design parameter. The total altitude control is finally defined as

$$T_z = T_{zeq} + T_{zconv}. \tag{26}$$

B. NTSMC-Based Attitude Control Design

The altitude control loop generates the required total thrust, whereas the attitude control subsystem regulates the quadrotor orientation in order to ensure the desired motion of the vehicle. The objective of the attitude control subsystem is to stabilize the quadrotor rotational dynamics and ensure tracking of the desired Euler angles $(\phi_{ref}, \theta_{ref}, \psi_{ref})$ by generating the control torques τ_x, τ_y and τ_z.

The attitude controller is composed of three independent NTSMC loops designed for the roll, pitch and yaw angles.

1) Roll Angle Controller Design: The roll motion corresponds to the rotation around the body x-axis. The roll tracking error is defined as

$$e_\phi = \phi - \phi_{ref}. \tag{27}$$

The corresponding nonsingular terminal sliding variable is selected as

$$\sigma_\phi = e_\phi + \frac{1}{c_{1\phi}} \dot{e}_\phi^{\frac{p}{q}}. \tag{28}$$

The equivalent control component is obtained by differentiating the sliding variable and imposing the sliding condition $\dot{\sigma}_\phi = 0$, i.e.

$$\dot{\sigma}_\phi = \dot{e}_\phi + \frac{1}{c_{1\phi}} \frac{p}{q} \dot{e}_\phi^{(\frac{p}{q}-1)} \ddot{e}_\phi = 0, \tag{29}$$

which leads to

$$\ddot{\phi} = \ddot{\phi}_{ref} - c_{1\phi} \frac{q}{p} \dot{e}_\phi^{(2-\frac{p}{q})}. \tag{30}$$

Differentiating the roll dynamics given in (16) yields

$$\ddot{\phi} = \dot{P} + \dot{Q}A_3 + Q\dot{A}_3 + \dot{R}A_4 + R\dot{A}_4, \tag{31}$$

where the auxiliary quantities $A_3 = s_\phi t_\theta$ and $A_4 = c_\phi t_\theta$ are defined to simplify the notation. Substituting the quadrotor dynamic equations into the sliding condition and setting (30) equal to (31), while also taking (13) into account, the equivalent control component can be written as

$$\begin{aligned} \tau_{xeq} = & I_{xx} \Big[\ddot{\phi}_{ref} - c_{1\phi} \frac{q}{p} \dot{e}_\phi^{(2-\frac{p}{q})} - a_1 QR + \\ & + a_2 QW_G - \dot{Q}A_3 - Q\dot{A}_3 - \dot{R}A_4 - \dot{A}_4 R \Big], \end{aligned} \tag{32}$$

where $a_1 = \frac{I_{yy}-I_{zz}}{I_{xx}}$ and $a_2 = \frac{I_{zzm}}{I_{xx}}$ are introduced to make the expression more compact.

To enforce convergence toward the sliding surface, the switching control component is introduced as

$$\tau_{xconv} = -c_{2\phi} \mathrm{sgn}(\sigma_\phi), \tag{33}$$

where $c_{2\phi} > 0$ is a design parameter. The total roll control torque is therefore given by

$$\tau_x = \tau_{xeq} + \tau_{xconv}. \tag{34}$$

2) Pitch and Yaw Controller Design: The objective of the pitch and yaw control subsystem is to ensure accurate tracking of the desired pitch and yaw angles θ_{ref} and ψ_{ref} by designing NTSMC-based control laws for the corresponding rotational dynamics. The control laws for the pitch and yaw channels are derived analogously to the roll controller by applying the same NTSMC design procedure. The resulting control torques are expressed as

$$\tau_y = \tau_{yeq} - c_{2\theta} \mathrm{sgn}(\sigma_\theta) \tag{35}$$

$$\tau_z = \tau_{zeq} - c_{2\psi} \mathrm{sgn}(\sigma_\psi), \tag{36}$$

with the equivalent control components designed as

supposed to be only 2 eq.

$$\tau_{yeq} = \frac{I_{yy}}{c_{1\theta}} \Big[\ddot{\theta}_{ref} - c_{1\theta} \frac{q}{p} \dot{e}_\theta^{(2-\frac{p}{q})} - b_1 PRc_\phi + \tag{37}$$

$$-b_2 PW_G c_\phi + PQs_\phi + \dot{R}s_\phi + PRc_\phi \Big], \tag{38}$$

$$\tau_{zeq} = \frac{I_{zz}}{A_6} \Big[\ddot{\psi}_{ref} - c_{1\psi} \frac{q}{p} \dot{e}_\psi^{(2-\frac{p}{q})} - Q\dot{A}_5 + \tag{39}$$

$$-\dot{Q}A_5 - R\dot{A}_6 - d_1 A_6 PQ \Big], \tag{40}$$

where $b_1 = \frac{I_{zz}-I_{xx}}{I_{yy}}$, $b_2 = \frac{I_{zzm}}{I_{yy}}$, $d_1 = \frac{I_{xx}-I_{yy}}{I_{zz}}$, $A_5 = \frac{s_\phi}{c_\theta}$, $A_6 = \frac{c_\phi}{c_\theta}$, while $c_{1\phi}, c_{2\phi}, c_{1\theta}$ and $c_{2\theta}$ represent positive controller design parameters.

3) Stability Analysis: The stability of the proposed NTSMC control scheme can be established using Lyapunov theory. Consider the Lyapunov candidate function

$$V = \frac{1}{2}\sigma_i^2, \tag{41}$$

where σ_i denotes the sliding variable associated with the corresponding controlled subsystem. Taking the time derivative of V yields

$$\dot{V} = \sigma_i \dot{\sigma}_i. \tag{42}$$

Substituting the control law into the sliding dynamics leads to

$$\dot{\sigma}_i = -c_{2i} \mathrm{sgn}(\sigma_i) + d_i, \tag{43}$$

where d_i represents a bounded disturbance. The disturbance bounds are assumed to be known or conservatively overestimated, which is a standard assumption in SMC design. Assuming that

$$|d_i| \le \Delta_i,$$

the derivative of the Lyapunov function satisfies

$$\dot{V} \le -(c_{2i} - \Delta_i)|\sigma_i|. \tag{44}$$

Therefore, if the gain is selected such that

$$c_{2i} = \Delta_i + \alpha, \ \alpha > 0,$$

the condition $\dot{V} < -\alpha\sqrt{V}$ is satisfied, which guarantees convergence of the sliding variables to zero in finite time. Consequently, the tracking errors of the attitude subsystem

converge to zero in finite time. This ensures that the closed-loop system trajectories reach the sliding surface in finite time and remain on it thereafter.

IV. SIMULATION RESULTS

This section presents simulation results obtained for the quadrotor UAV system under realistic disturbance-affected operating conditions. The objective is to evaluate the performance, robustness, and practical implementation aspects of the proposed control strategy. In particular, the proposed *NTSMC* controller is compared with *FOSMC* [9] and *SOSMC* [14] under disturbance-affected operating conditions. To provide a comprehensive assessment, three simulation scenarios are considered. The first scenario investigates the practical implementation aspect of the controller by analyzing chattering reduction using a HGA mechanism [14]. The second scenario presents a comparative evaluation of NTSMC with conventional sliding mode approaches under persistent disturbances. Finally, the third scenario examines controller robustness during aggressive maneuvers and strong disturbance conditions.

All simulations were carried out in MATLAB/Simulink using a fixed-step solver with a sampling time of $T_s = 1\,\mathrm{ms}$, corresponding to a control update frequency of $1\,\mathrm{kHz}$. This setting is representative of real-time quadrotor control systems and is suitable for digital implementation. The system parameters, as well as the FOSMC and SOSMC controller parameters, were directly adopted from [9] and [14], respectively, while the NTSMC parameters were selected as $c_1 = [80,\ 10,\ 10,\ 60]$ and $c_2 = [225,\ 20,\ 20,\ 1]$ for the channels $[z,\ \phi,\ \theta,\ \psi]$, with $p = 5$ and $q = 3$. The controller gains were tuned using a trial-and-error procedure with emphasis on ensuring robust closed-loop stability under disturbances, rather than on fine-tuning the transient response. Consequently, the reported results reflect the intrinsic robustness properties of the proposed control strategy rather than performance obtained through aggressive parameter optimization. good, practical approach

A. Scenario 1 – Chattering Reduction Using HGA mechanism

Controllers relying on discontinuous switching actions inherently exhibit high-frequency chattering in both system states and control input. While such switching enhances robustness from a theoretical standpoint, it is undesirable in practical implementations due to increased actuator activity and potential degradation of control performance. To address this limitation, a high-gain adaptation (HGA) strategy is incorporated into the NTSMC framework to attenuate the switching intensity while preserving the essential robustness properties of the controller.

In this scenario, the Vivian curve is selected as the reference trajectory, with the perturbed initial condition $z_{ref}(0) = 1\,\mathrm{m}$, while external disturbances are neglected to isolate the effect of the chattering-reduction mechanism. The HGA algorithm is activated within a small neighborhood of the sliding manifold, where the discontinuous control

plotting the control error is more meaningful

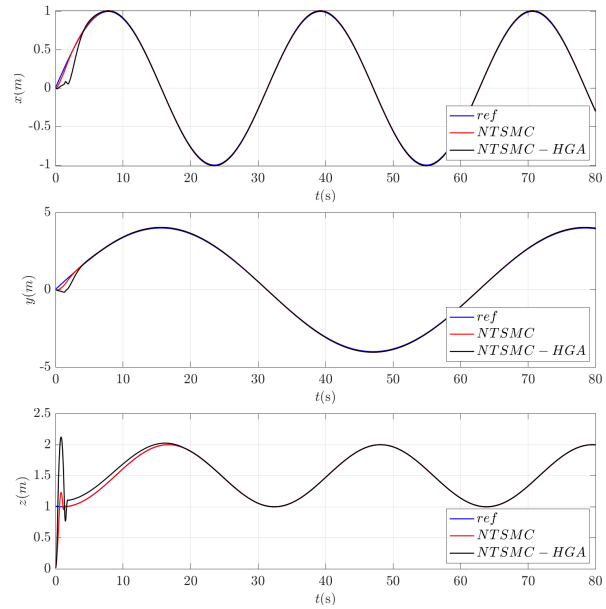


Fig. 2: Time responses of the quadrotor UAV position (Scenario 1)

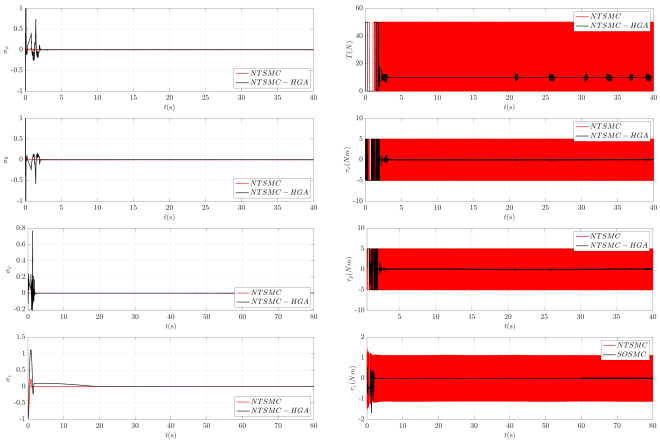
term is replaced by a filtered counterpart using the low-pass transfer function $G_{NPF}(s) = \frac{K}{Ts+1}$. The parameters of the HGA mechanism are directly adopted from [14] without modification. The obtained results clearly illustrate the expected trade-off. As shown in Fig. 2 and Fig. 3 (a), the original NTSMC controller achieves slightly better tracking performance, while NTSMC-HGA introduces modest transient deviations and less accurate convergence toward the sliding manifold. In contrast, the control inputs depicted in Fig. 3 (b) demonstrate a significant reduction in high-frequency switching when the HGA mechanism is applied. The control signals become considerably smoother after the transient phase, indicating effective chattering attenuation.

Therefore, although the HGA-based modification reduces chattering, this improvement is achieved at the expense of tracking accuracy and robustness. Consequently, the original NTSMC controller is retained in subsequent scenarios, where robustness to disturbances is of primary interest.

B. Scenario 2 - Robustness properties and tracking performance of FOSMC, SOSMC and NTSMC techniques under non-vanishing disturbances

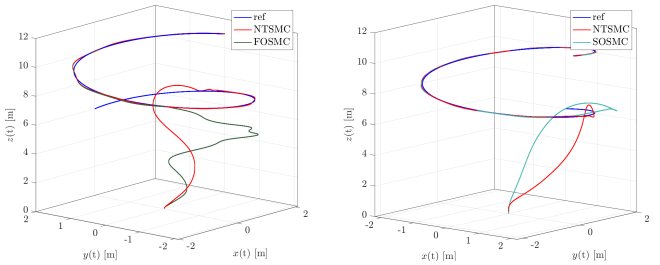
In this scenario, the FOSMC, SOSMC, and NTSMC algorithms are evaluated under identical perturbed initial conditions, with $y_{ref}(0) = 2$ and $z_{ref}(0) = 6$, and in the presence of three non-vanishing matched disturbances $d_{\tau_x}(t) = d_{\tau_y}(t) = d_{\tau_z}(t) = \sin(3t)$, as well as two non-vanishing unmatched disturbances defined as $d_\phi(t) = d_\theta(t) = \sin(3t)$ acting through channels (16) and (17). The objective is to assess the robustness of the controllers under simultaneous perturbations and external disturbances.

As shown in Fig. 4 and Fig. 5, the comparison between the controllers highlights different dynamic characteristics. FOSMC provides smoother lateral tracking during the tran-



(a) Sliding variables (b) Control inputs

Fig. 3: Trade-off between control performance and chattering attenuation using HGA mechanism (Scenario 1)



(a) (b)

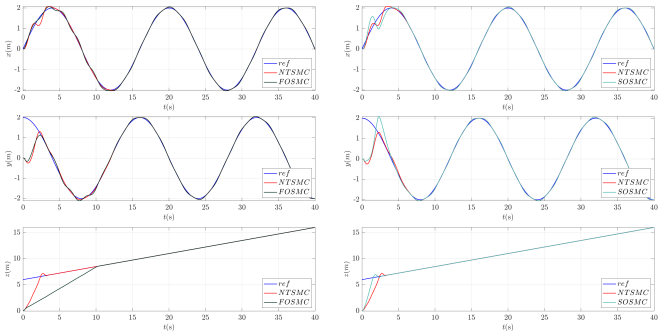
Fig. 4: 3D trajectory tracking using the FOSMC, SOSMC and NTSMC under matched and unmatched non-vanishing disturbances (Scenario 2)

sient phase, while NTSMC achieves faster and more stable altitude convergence (see Fig. 4 (a) and Fig. 5 (a)). In comparison with SOSMC, both controllers maintain convergence; however, NTSMC demonstrates smaller oscillations in the lateral motion, while SOSMC improves altitude tracking accuracy during the transient (see Fig. 4 (b) and Fig. 5 (b)).

The results obtained in this scenario indicate that the SOSMC and the proposed NTSMC provide the most balanced performance in terms of convergence speed, tracking accuracy, and disturbance rejection. However, in order to further evaluate the robustness of the control strategies under more demanding operating conditions, an additional scenario involving aggressive maneuvers and persistent disturbances is considered in the following subsection.

C. Scenario 3 – *Robustness under Aggressive Maneuvers and Persistent Disturbances*

In the final scenario, the comparison is continued between NTSMC and SOSMC only, since SOSMC demonstrated stronger performance than FOSMC in the previous study and therefore represents a more demanding benchmark. The system is subjected to persistent matched disturbances acting on τ_x, τ_y, and τ_z, together with unmatched disturbances acting through ϕ and θ, all defined as $d(t) = \sin(3t)$. The



(a) (b) NTSMC with SOSMC

Fig. 5: Position of the quadrotor UAV using the FOSMC, SOSMC and NTSMC under matched and unmatched non-vanishing disturbances (Scenario 2)

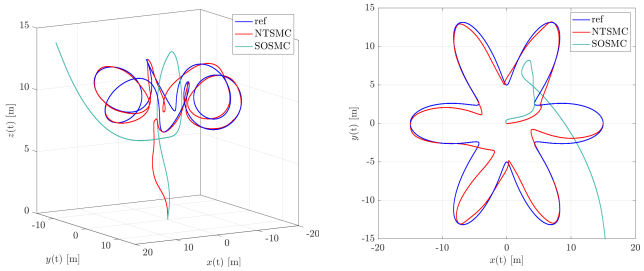


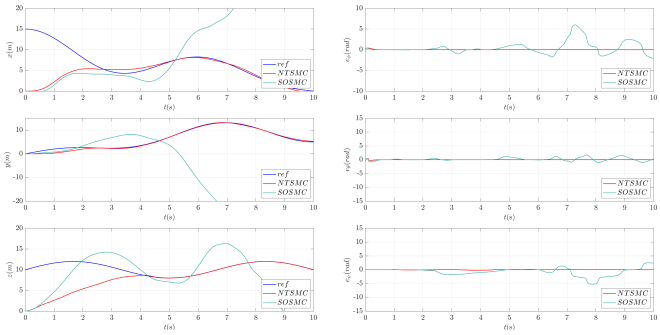
Fig. 6: Quadrotor UAV tracking of 3D trajectory with (x, y) planar projection using NTSMC and SOSMC under matched and unmatched non-vanishing disturbances and aggressive maneuvers (Scenario 3)

scenario was selected to make tracking in the x-y plane more difficult and can therefore be viewed as a representative case of lateral wind-like disturbance. The results presented in Figs. 6–7 show a clear difference between the controllers. While SOSMC gradually loses convergence and deviates from the desired trajectory, NTSMC maintains bounded and stable tracking throughout the simulation. Although SOSMC reaches the desired altitude slightly faster, this response is accompanied by significant overshoot, indicating aggressive control action. Consequently, the proposed NTSMC controller demonstrates superior robustness and more reliable trajectory tracking under aggressive operating conditions and strong disturbances.

V. CONCLUSIONS

This paper has presented a nonsingular terminal sliding mode control (NTSMC) strategy for robust three-dimensional trajectory tracking of a quadrotor UAV in the presence of matched and unmatched disturbances. The proposed control scheme ensures finite-time convergence of the sliding variables and tracking errors while avoiding the singularity issues typically associated with classical terminal sliding mode control formulations. The main contribution of this work lies in the development and systematic evaluation of a NTSMC-based control framework that achieves a favorable trade-off between robustness, convergence speed, and tracking accu-

thicker lines is recommended



(a) (b)

Fig. 7: Position coordinates and orientation tracking errors of the quadrotor UAV using NTSMC and SOSMC under matched and unmatched non-vanishing disturbances and aggressive maneuvers (Scenario 3)

racy in disturbed operating conditions. In particular, the proposed controller demonstrates improved altitude control and more balanced overall performance compared to FOSMC and SOSMC approaches. The conducted stability analysis confirms finite-time convergence of the sliding variables and tracking errors under bounded disturbances, while the simulation results validate the theoretical findings through comparative studies under increasingly demanding scenarios. In contrast to SOSMC, which may exhibit faster but less stable responses, and FOSMC, which provides smoother but slower convergence, the NTSMC approach achieves the most consistent closed-loop behavior across all considered conditions. In addition, a high-gain adaptation (HGA) mechanism has been incorporated to mitigate chattering effects inherent to discontinuous control laws. The obtained results highlight the fundamental trade-off between chattering reduction and robustness, showing that improved control smoothness is achieved at the expense of reduced disturbance rejection capability.

Although the proposed approach demonstrates strong robustness and favorable transient performance, its effectiveness depends on the availability of accurate state measurements and reasonable estimates of disturbance bounds. Future work will therefore focus on experimental validation and the development of adaptive NTSMC framework to further improve robustness and reduce chattering in real-world applications.

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